

Hacking in the U.S. Healthcare

The Target: The Financial Backbone of Healthcare

Imagine a single company that handles the payments and insurance claims for one out of every three Americans. That's Change Healthcare. It's the financial plumbing of the U.S. healthcare system, connecting doctors, hospitals, and pharmacies. This central role, while efficient, also made it a single point of failure.

The Attack: A Door Left Unlocked

In February 2024, the notorious ransomware group BlackCat launched a calculated attack. They weren't just looking for any target; they were retaliating against the FBI for a previous crackdown and wanted to cause maximum disruption.

Their way in wasn't a brilliant, complex hack. It was shockingly simple. They found a remote access portal—a digital door into the company's network—that wasn't protected with multi-factor authentication (MFA). Using stolen credentials, they just walked right in. For nine days, they moved silently through the network, stealing a massive 6 terabytes of data before deploying their ransomware, which encrypted everything and brought the company to a grinding halt.

The Aftermath: A National Crisis

The impact was immediate and devastating. Pharmacies couldn't verify insurance, forcing patients to pay out-of-pocket for vital medications. Hospitals and clinics couldn't get paid, pushing many smaller practices to the brink of collapse. The attack triggered a system-wide meltdown, forcing a return to paper forms and phone calls, and causing chaos for providers and patients alike.

The data breach was the largest in U.S. healthcare history, exposing the personal and medical information of nearly 193 million people.

The Resolution: A Painful Recovery

To stop the bleeding, Change Healthcare's parent company, UnitedHealth Group, paid a \$22 million ransom. They also had to pump over \$6 billion in emergency loans to struggling healthcare providers. The technical recovery was a monumental task that involved completely rebuilding their core systems from scratch. The total cost of the incident is expected to top \$2.4 billion, and the company is now facing dozens of class-action lawsuits and a federal investigation. It was a brutal lesson in what happens when a single, basic security control is

missed.

Snowflake Cloud Hack

The Target: A Giant Digital Warehouse

Snowflake is a massive cloud platform where thousands of major companies store and analyze their most important data. It's like a high-tech, secure warehouse for corporate information.

The Attack: Using Old Keys to Open New Doors

In mid-2024, a wave of data breaches hit some of Snowflake's biggest customers, including Ticketmaster and Santander Bank. At first, everyone assumed Snowflake itself had been hacked. But the truth was more complicated.

The investigation revealed that Snowflake's own systems were secure. The problem was with its customers. A hacking group known as Scattered Spider had been collecting passwords for years using "infostealer" malware—nasty software that steals saved logins from infected computers. Armed with a huge list of old, stolen credentials, the hackers started trying them on their owners' Snowflake accounts.

It worked. Why? Because hundreds of accounts belonging to major companies weren't protected by multi-factor authentication. The hackers used old, stolen keys to walk right into these companies' private data vaults on Snowflake and steal everything they could find.

The Aftermath: A Cascade of Breaches

The result was a disaster for the companies involved. Ticketmaster had the data of 560 million customers stolen and put up for sale online. Santander Bank lost the data of 30 million customers and staff. Dozens of other companies were hit, with the hackers demanding ransoms of up to \$5 million.

The Resolution: A Shared Responsibility

The incident was a wake-up call. Snowflake worked with cybersecurity firms to notify all affected customers and gave them a simple, urgent to-do list: turn on MFA, restrict access to trusted networks, and change all old passwords. The investigation also led to the arrest of key members of the hacking group.

This event redefined the idea of "shared responsibility" in the cloud. While a provider secures the platform, customers are responsible for securing their own front doors.

Silent Spies in the Phone Lines

The Target: The Heart of the U.S. Communications

This incident wasn't about money; it was a classic spy story. A highly sophisticated hacking group linked to the Chinese government, known as Salt Typhoon, targeted the core of America's telecommunications infrastructure. Their goal was simple: espionage.

The Attack: Exploiting Old, Unpatched Holes

Like the other attacks, Salt Typhoon's entry method wasn't a secret zero-day exploit. They got in by exploiting well-known security flaws in network equipment like routers and VPNs—flaws for which fixes had been available for years. They targeted major providers like AT&T and Verizon, patiently working their way deep inside the networks.

Once inside, they used custom, stealthy malware to hide their presence and maintain access for years. They "lived off the land," using standard administrative tools to move around so their activity would blend in with normal network traffic, making them nearly impossible to detect.

The Aftermath: A Major Intelligence Breach

The most alarming discovery was that the hackers had compromised the lawful intercept systems—the very tools the FBI uses for court-ordered wiretaps. They stole call and text message records and, in some cases, the actual audio of phone calls involving high-profile political targets.

This was a massive intelligence failure, giving a foreign adversary a direct window into U.S. government and political activities. The campaign went on for at least two years before it was discovered.

The Resolution: An Ongoing Hunt

The response has been a massive, ongoing effort led by the FBI, NSA, and CISA. Eradicating a deeply embedded state-sponsored actor from critical national infrastructure is an incredibly difficult and long-term challenge. Government agencies have issued urgent warnings, and the FCC is pushing for new rules to force telecom companies to practice basic cybersecurity hygiene. The hunt for these digital spies is far from over.